

TrustVote: Online Voting using BlockChain

1.Problem Statement :

Traditional voting systems often suffer from issues such as voter fraud, lack of transparency, and slow results, which can undermine public trust and participation. A blockchain-based e-voting system offers a solution by providing a secure, transparent, and tamper-proof platform for elections. This project aims to develop a blockchain e-voting system to enhance the integrity and efficiency of the voting process.

2.Abstract:

This project report presents the design and implementation of a Decentralized voting system using blockchain technology. The purpose of this system is to provide a secure, transparent, and immutable platform for conducting elections. The system employs the use of smart contracts on the Ethereum blockchain to automate the voting process and ensure that all votes are counted accurately. The system was developed using Solidity, a programming language for smart contracts, and Truffle, a development framework for Ethereum. The system's frontend was developed using HTML, CSS and JavaScript. Nodejs ,Expressjs and Mongodb for system's backend. The system's security was ensured through the use of encryption techniques and private keys to secure users' identities and their voting records. The system's architecture allows for the registration of voters, the creation of ballots, the casting of votes, and the tallying of results, all on the blockchain. The system's decentralized nature ensures that votes cannot be tampered with, and the results are transparent and available for all participants to verify. This project report concludes that a decentralized voting system using blockchain technology has the potential to revolutionize the way elections are conducted, by providing a secure, transparent, and immutable platform that can be trusted by all participants. The system can be used in various settings, including governmental and non-governmental elections, corporate boardroom elections, and more.

Keywords: Ethereum, Blockchain voting, Decentralization, Smart-contract

3.Introduction:



Democratic voting is such an important and significant event in every country, and the current way we vote usually involves either ballot papers or EVMs. While these methods are in place, they do come with some challenges, like

transparency issues, lower voter participation, vote tampering, a lack of trust in the election authority, problems with unique voter IDs, delays in announcing results, and, most importantly, security concerns. When we think about implementing a digital voting system, the security of digital voting really stands out as a primary concern. Given the enormous decisions involved, we want to be absolutely sure that the system can protect data and guard against possible attacks. One potential solution to these security challenges could be the use of blockchain technology. Blockchain technology opens up countless possibilities. It's a distributed ledger technology that enables digital assets to be exchanged in a peer-to-peer decentralized network. This advancement in distributed ledger technology is truly exciting! A block is essentially a collection of all the transactions that have taken place. Blockchain comes with fantastic features like immutability, decentralization, security, transparency, and anonymity. Combining blockchain with smart contracts presents a wonderful opportunity for creating a safer, more secure, and transparent e-voting system. In this paper, we've worked on and tested a sample e-voting application as a smart contract for the Ethereum network, utilizing blockchain technology through wallets and the Solidity programming language.

4. Literature Survey (comparison with existing system):

Id	Year,Title,Author	Methodology	Limitation	Comparison to Current Study
[1]	January 2018 Decentralized E-Voting Systems Based on the Blockchain Technology Jen-Ho Hsiao ¹ , Raylin Tso ¹ , Chien-Ming Chen ² and Mu-En Wu ³	Blockchain Integration: The system uses a blockchain ledger to ensure the immutability and transparency of votes, providing a tamper-proof record of all voting transactions. Smart Contracts: Deployed on the blockchain, smart contracts automate the voting process, including ballot counting and result verification, replacing traditional bulletin boards and ensuring accurate vote tallying. Secret Sharing Scheme: Utilizes Shamir's secret sharing to protect	Scalability Issues: Performance Concerns: Blockchain systems can face performance challenges, particularly with high transaction volumes. The proposed system may experience delays or inefficiencies when scaled to handle large numbers of voters and candidates.	

		vote confidentiality by dividing votes into shares distributed among multiple servers, preventing unauthorized access. Roles and Responsibilities include: Voters (Vi): Eligible individuals who receive voting certificates. Registration Server (RS): Verifies identities and issues certificates. Authentication Server (AS): Validates the certificates issued by RS. Voting Website (VWeb): Online platform for voting, managed by electoral authorities.	Complexity of Implementation: Technical Complexity: Integrating blockchain, smart contracts, and advanced cryptographic techniques requires significant technical expertise. Privacy Concerns: While the system aims to protect voter anonymity through encryption and secret sharing, concerns about potential privacy breaches or unauthorized data access persist. The robustness of the encryption methods and their implementation must be continually evaluated.	
[2]	ICMPC 2023 Decentralized and Automated Online Voting System using Blockchain Technology Polepaka Sanjeeva1*, M. Sai Sathwik1, G. SaiPrasad1.	Blockchain Integration: The proposed online voting system uses Ethereum Blockchain to provide a decentralized ledger for recording votes. Smart Contracts: Smart contracts on the Ethereum Blockchain automate the voting process, including ballot counting and result verification. Facial Recognition: Voter authentication is performed through facial recognition	1.Facial Recognition Accuracy: Facial recognition systems may have limitations in terms of accuracy and reliability, particularly under varying lighting conditions or with diverse demographic groups. 2.Manipulation Risks: Online voting systems may be susceptible to coercion or vote	

		technology. This biometric verification ensures that only eligible voters can participate and prevents fraudulent voting by verifying voter identity against stored data. Voting Process: Voters access the system, undergo facial recognition for authentication, and cast their votes. Successful authentication allows voters to select candidates from a district-specific list. Vote Verification: Votes are verified by multiple miners who ensure that only valid votes are added to the public ledger. Each vote is processed as a separate transaction, and malicious votes are rejected. Transaction Tracking: Voters receive notifications with transaction IDs for tracking their votes. This system ensures that voters can verify their participation while maintaining confidentiality.	buying. Designing mechanisms to detect and prevent these practices is essential to maintain the fairness of elections. 3.Data Exposure: While Blockchain provides transparency, it can also expose voter identities and preferences. Balancing transparency with privacy is crucial, and techniques like zero-knowledge proofs or advanced encryption methods may be required to address this issue.	
[3]	2020 A Secure Digital Voting System based on Blockchain Technology S. S. Bhalerao, et al.	Proposed a secure digital voting system using blockchain technology, emphasizing the system's architecture, security features, and implementation details.	Focused primarily on the technical aspects, with limited discussion on scalability and user accessibility.	This study builds on the concept by addressing additional aspects such as scalability and practical deployment in large-scale elections.

[4]	2018 Blockchain-Based E-Voting System A. Kiayias, et al.	Implemented a blockchain-based e-voting system, focusing on privacy, verifiability, and resistance to tampering.	Lacked practical deployment analysis and did not consider large-scale election scenarios.	The current study compares the implementation strategies and enhances the security protocols for better real-world applicability.
[5]	June 2016 Electronic Voting Service Using Block-Chain Kibin Lee, Joshua I. James, Tekachew Gobena Ejeta, Hyoung Joong Kim	The paper introduces a novel voting system that leverages blockchain technology to enhance security and transparency. It involves several steps: filtering transactions, implementing a consensus protocol, and using a decentralized ledger to record votes. The system incorporates a trusted third party (TTP) for additional validation and employs blockchain for immutability.	The system's effectiveness depends on the accurate functioning of the TTP and the blockchain protocol. Challenges include potential vulnerabilities in the transaction filtering process, maintaining consensus in a decentralized network, and addressing scalability issues. Additionally, the system's dependence on the TTP could introduce points of failure or bottlenecks.	This study Maintains redundant copies of the blockchain data to prevent data loss and ensure continuity in case of an attack or failure.
[6]	2019 Blockchain And Finger Print Enabled E-Voting Aravind , Gokul Raj ,Mohanraj	Proposed system Fingerprint Authentication for Identity Verification due to faster and better than other biometric data.They have used Blockchain Data Storage and Homomorphic encryption. Also the system has two types of users one is Election Officer &	The paper mentions that no full implementation of blockchain-based e-voting has occurred for a national election. Although the proposed system aims to reduce costs associated with	The proposed system aims to decentralize the voting process, potentially increasing security and transparency.

		another is Booth Manager, Booth Manager System developed with voter's functionality where voters are going to poll.	traditional voting methods, the complexity of implementing this may present significant challenges.	
[7]	2023 Transparent voting system using blockchain V. Anitha ,*, Orlando Juan Marquez Caro , R. Sudharsan S. Yoganandan M. Vimal	The methodology of the research paper involves creating a blockchain-based voting system to ensure secure and transparent elections. It utilizes the Ethereum blockchain and smart contracts to automate the voting process, including voter authentication and vote tallying, eliminating the need for intermediaries. The system ensures voter anonymity through cryptographic techniques and provides real-time election results to enhance transparency.	Complex for Non-Tech Users: Users unfamiliar with blockchain may find it difficult to use digital wallets like MetaMask. Scalability Issues: The system might slow down with a large number of votes due to Ethereum's transaction limits. Cost of Transactions: Each vote incurs a fee on the Ethereum network, which could be costly in large elections.	The current study compares the implementation of the existing blockchain voting system comprising three levels: authentication, voting, and counting.
[8]	June 2024 Online Voting Platform Using Blockchain Technologies Sonali Rangdale, Nagesh Raykar, Santosh Borde, Prashant Kumbharkar, Udit kumar	The study describes a blockchain-based electronic voting system in which the system uses homomorphic encryption and the cryptographic hash algorithm Keccak-256 to enable voter identity verification, safe ballot issuance, and encrypted vote storage. Voters may monitor and verify their ballots while preserving data integrity and privacy thanks to the decentralized Ethereum blockchain used to store votes. The most recent votes are	The system's scalability is hampered by the difficulty of handling many votes at once, particularly in areas with big populations. Additionally, its performance and accessibility may be restricted by the requirement for technical know-how, substantial device	By employing more effective consensus techniques, streamlining the blockchain for increased performance, and developing user-friendly interfaces with offline voting choices for places with inadequate internet connectivity, we may get around these

		counted, and the mechanism is meant to avoid duplicate voting.	resources, and reliable internet connectivity.	problems. Furthermore, increasing technological accessibility can be achieved through user education.
[9]				
[10]	July 2018 On Trade-offs of Applying Block Chains for Electronic Voting Bulletin Boards Sven Heiberg ¹ , Ivo Kubjas ¹ , Janno Siim ^{3,4} , and Jan Willemson ^{2,3}	The authors highlight that existing public blockchains like Bitcoin and Ethereum have significant drawbacks and argue that more in-depth research is needed to create a practical and reliable blockchain voting system.	-	-

5. Methodology / algorithms / techniques used :

Voter Registration:

- Input: Voter submits registration form with Aadhar number and other credentials.
- Process:
 - Validate the credentials.
 - Record voter information on the blockchain (create a blockchain entry).
- Output: Registration confirmation.

Voter Login:

- Input: Voter enters Aadhar number and password.
- Process:
 - Verify the credentials using the backend (Node.js, Express.js, MongoDB).
 - Grant access if valid.
- Output: Access to the dashboard.

View Candidates:

- Input: Voter selects the "View Candidates" option.
- Process:
 - Fetch the list of candidates from the blockchain.
- Output: Display candidates to the voter.

Cast Vote:

- Input: Voter selects a candidate and submits the vote.
- Process:
 - Verify voter eligibility (one vote per account).
 - Record the vote on the blockchain using a smart contract.
 - Send email verification to the voter.
- Output: Vote confirmation and record on blockchain.

View Results (User/Admin):

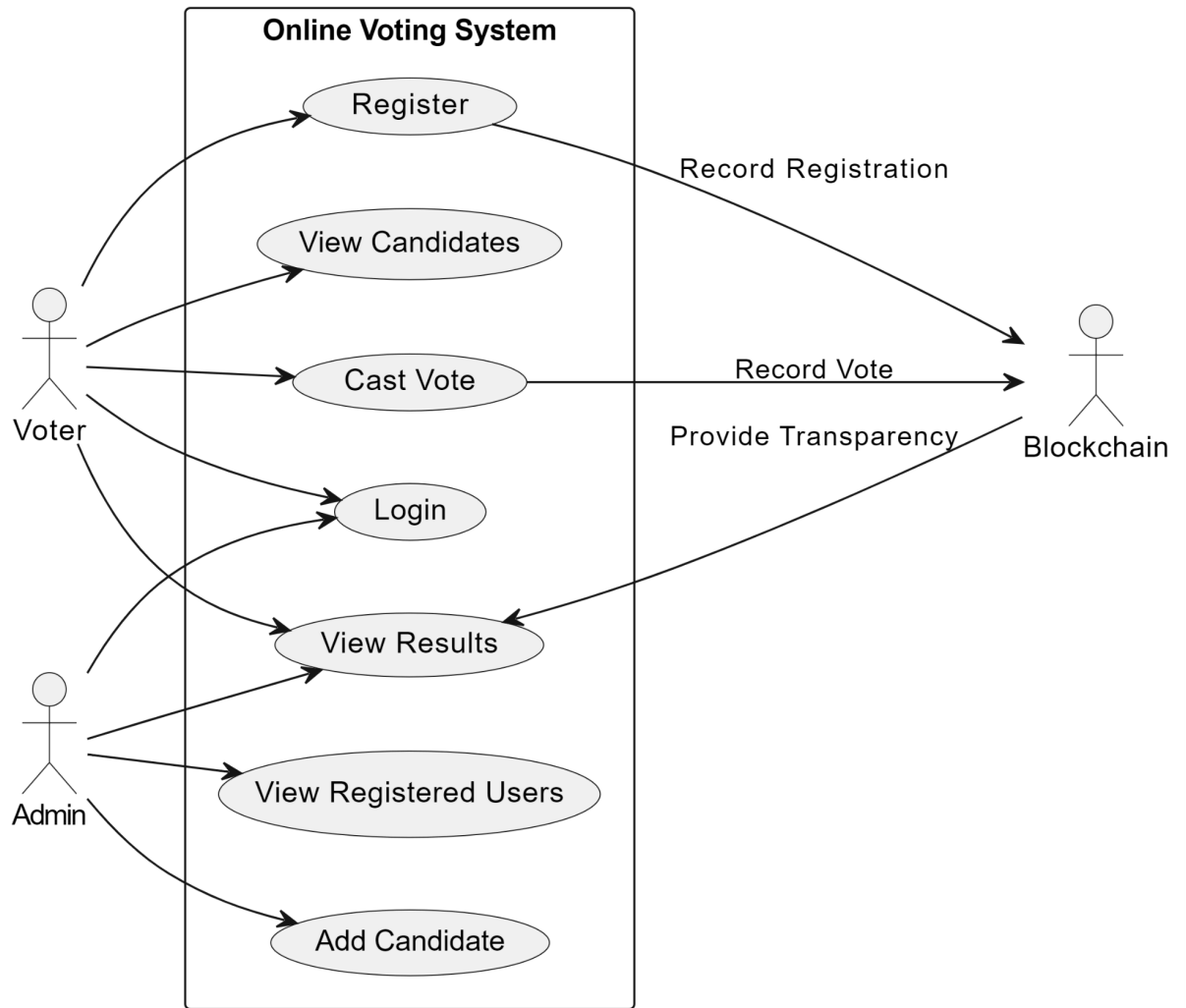
- Input: User or admin selects "View Results" option.
- Process:
 - Fetch the voting results directly from the blockchain.
- Output: Display transparent, immutable results.

Admin Functions:

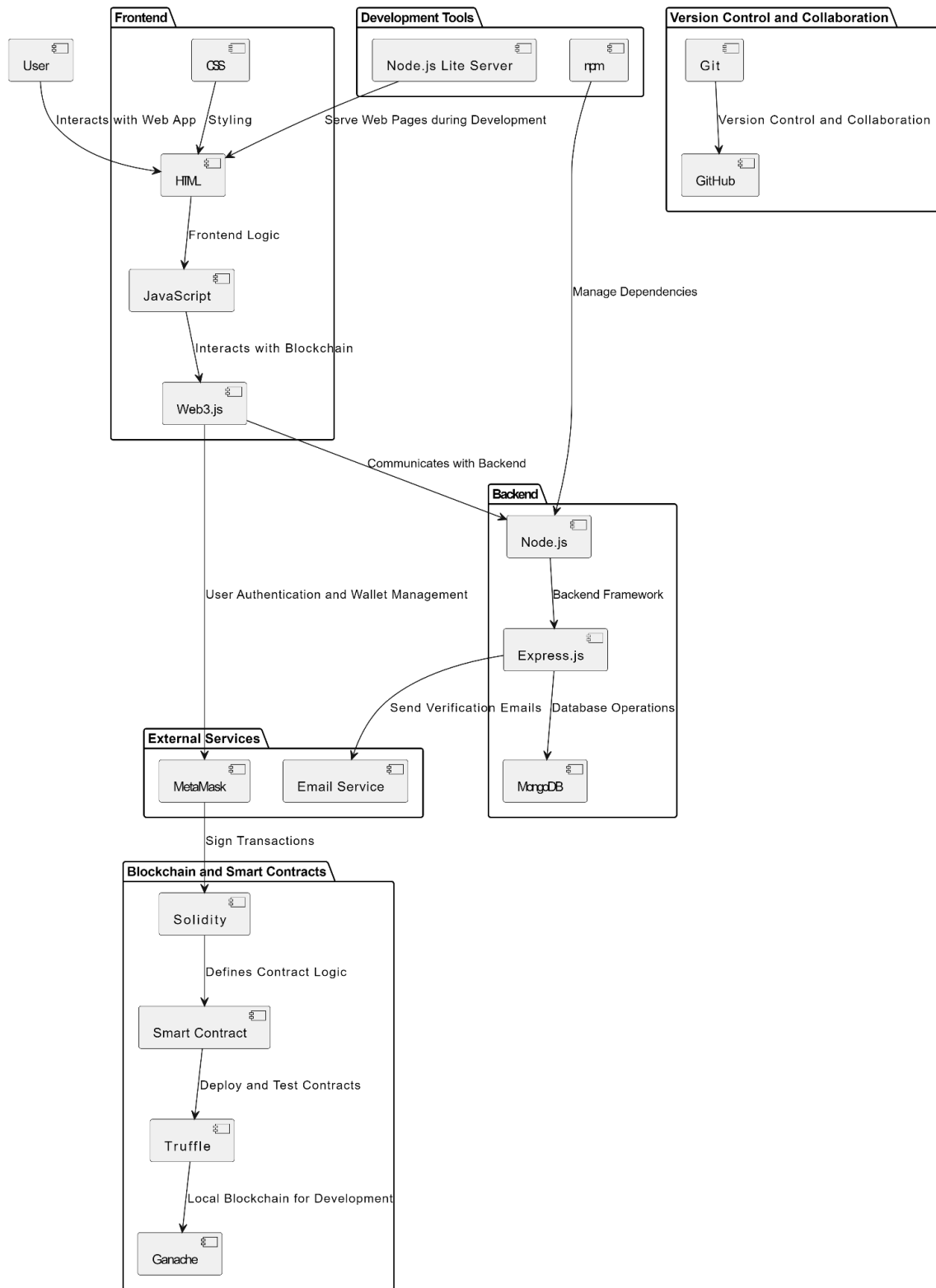
- **Add Candidates:** Admin enters candidate details and submits.
 - Process: Record new candidate entry in the blockchain.
- **View Registered Users:** Fetch list of registered voters from the database.
- **View Results:** Access voting results.

6. Design : UML diagrams:

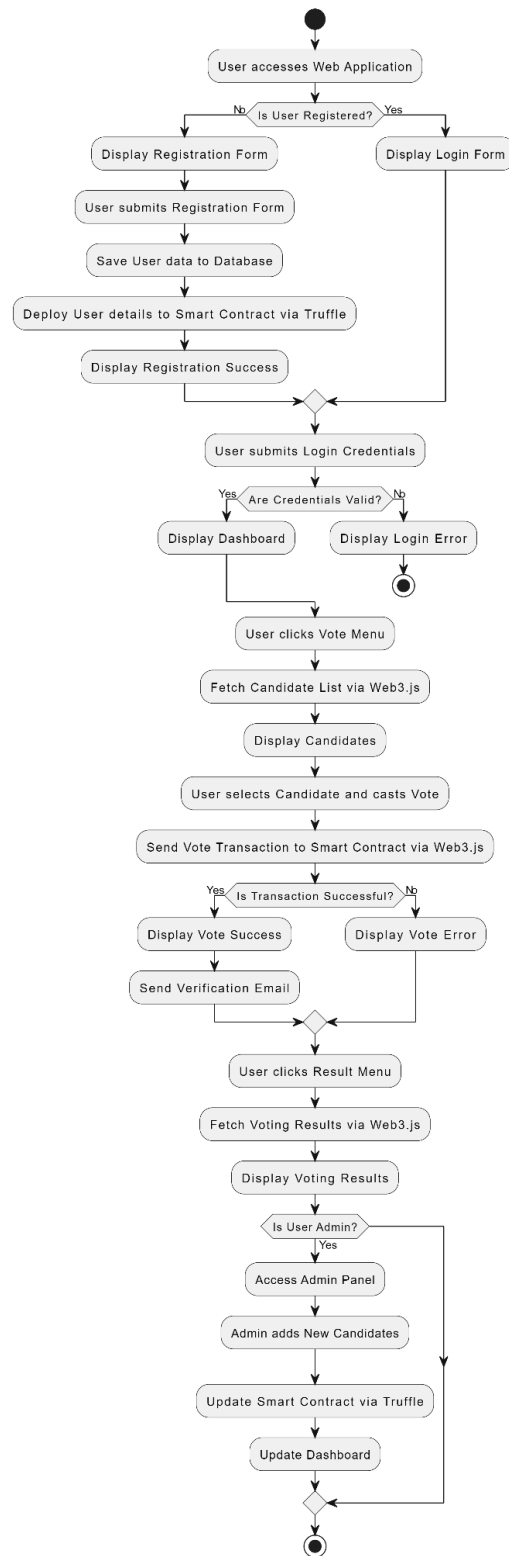
1. Use Case Diagram



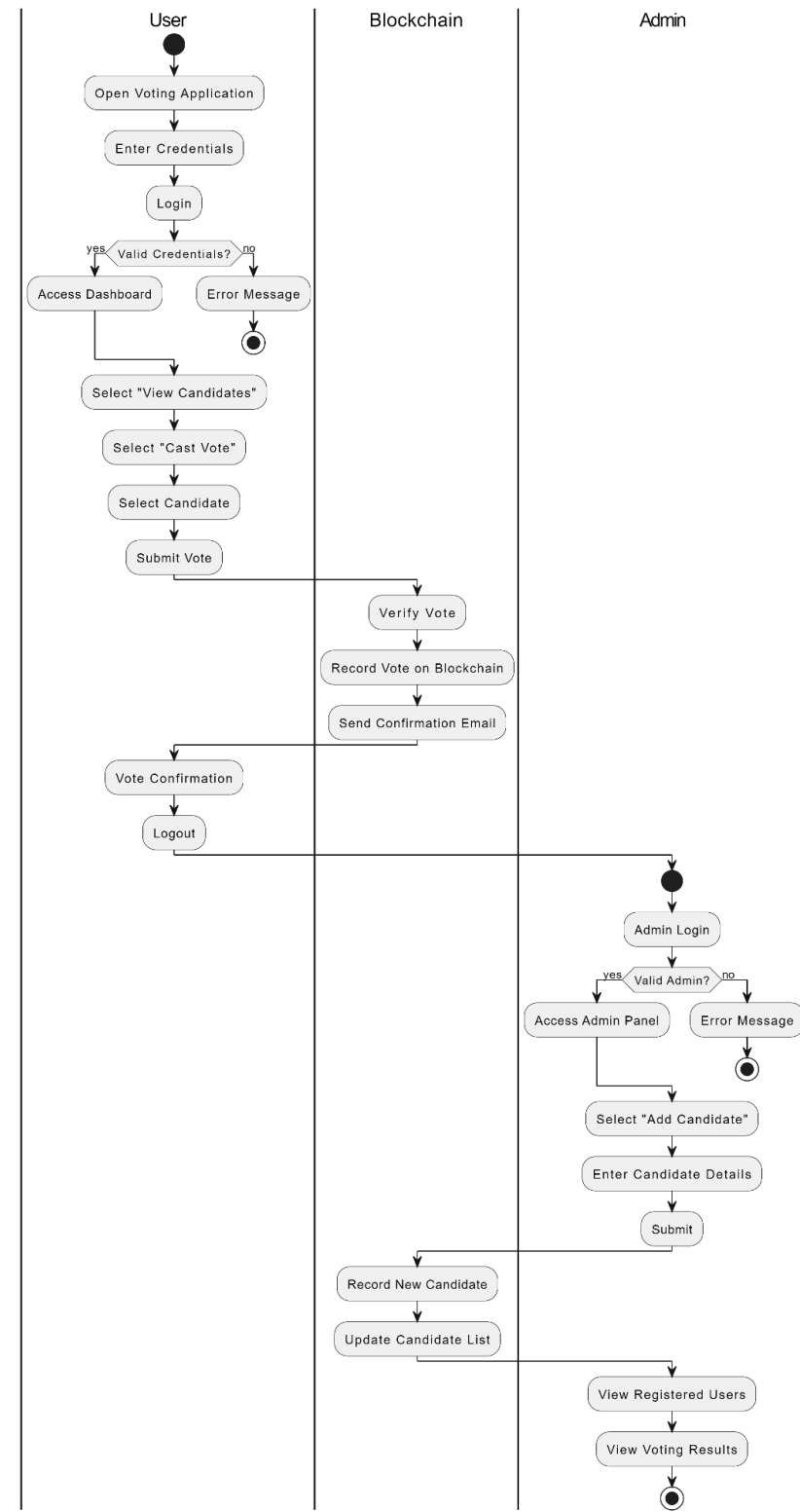
2.Block Diagram



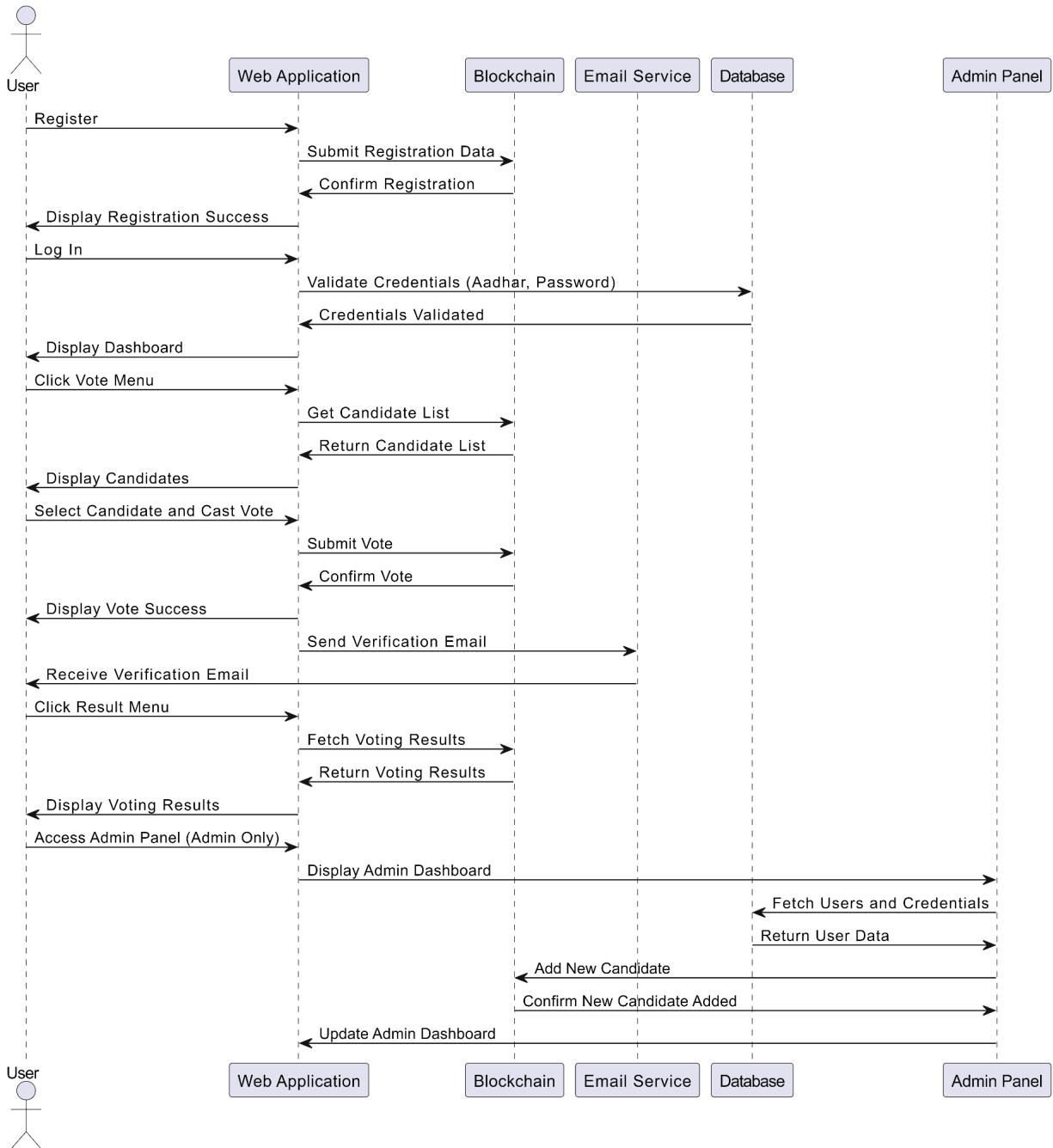
3. Flow Diagram



4. Activity Diagram



5. Sequence Diagram



7. Proposed System:

1. In this system the voter/user must first register themselves using a registration form available within the web application and once the registration form is being submitted, an entry is made in the blockchain.
2. After the registration, the user can log into the application and be a part of the polling process. The user with valid credentials can log into the system by entering their Aadhar number and password.
3. Once the user is logged into their respective account the dashboard contains all the information about voting.
4. When users click on the vote menu, they will see the candidates of the respective parties.
5. Users will select one party and cast the vote for that party. Only one vote is casted from each account.
6. Once the vote is cast an Email verification will be sent to the registered account and that account is registered into the blockchain.
7. The system has a secure login system which prevents people from casting votes on behalf of others.
8. By clicking the result on the menu ,the user can see the result on the dashboard.
9. Dashboard also contains an admin panel only accessed by the admin; the admin can add new candidates into the system and see the user's name with its valid credentials.

8. Software Tools / Technologies to be used:

1. Frontend Development

- **HTML (HyperText Markup Language):** The standard markup language used for creating web pages and structuring content on the web.
- **CSS (Cascading Style Sheets):** A style sheet language used for describing the presentation of a document written in HTML. CSS is used to control the layout, colors, fonts, and overall appearance of the web pages.
- **JavaScript:** A versatile programming language primarily used to create dynamic and interactive effects within web browsers.
- **Web3.js:** A JavaScript library that allows interaction with the Ethereum blockchain, enabling you to send transactions, interact with smart contracts, and retrieve user accounts.
- **Node.js Lite Server:** A lightweight Node.js server used to serve web pages during development, providing live-reload capabilities for a better development experience.

2. Backend Development

- **Node.js:** A JavaScript runtime built on Chrome's V8 JavaScript engine, used for developing fast and scalable server-side applications. It allows for server-side programming using JavaScript.
- **Express.js:** A minimal and flexible Node.js web application framework that provides a robust set of features to build single-page, multi-page, and hybrid web applications. It's commonly used for building RESTful APIs.
- **MongoDB:** A NoSQL database program that uses JSON-like documents with optional schemas. MongoDB is known for its scalability, flexibility, and ease of use, making it suitable for handling large volumes of data.

3. Blockchain and Smart Contracts

- **Solidity:** A statically typed, contract-oriented programming language designed for developing smart contracts that run on the Ethereum blockchain. Solidity is used to write the smart contracts that define the logic of your voting system.
- **Truffle:** A development environment, testing framework, and asset pipeline for Ethereum. It facilitates the development, deployment, and testing of smart contracts using the Ethereum blockchain.

4. Version Control and Collaboration

- **Git:** A distributed version control system used for tracking changes in source code during software development. It allows multiple developers to collaborate on a project efficiently.
- **GitHub:** Cloud-based Git repository hosting services that provide additional features like issue tracking, continuous integration, and project management tools.

5. Development and Deployment Tools

- **Ganache:** A personal blockchain for Ethereum development that you can use to deploy contracts, develop your applications, and run tests.
- **MetaMask:** A browser extension that allows users to interact with Ethereum-based DApps directly from their web browsers by providing a secure identity vault and user interface.
- **npm (Node Package Manager):** A package manager for JavaScript that comes with Node.js. It is used to install and manage dependencies for your project.

9. Proposed Outcomes:

The screenshot shows the 'Create an account' registration form on the BLOCKCHAIN-VOTE website. The form is set against a dark blue background with a network diagram pattern. At the top, the website name 'BLOCKCHAIN-VOTE' is on the left, and navigation links 'Register', 'Login', 'About', 'Results', and 'Admin Panel' are on the right. The main heading 'Create an account' is centered. The form fields include: 'First Name' (Shrinivas), 'Last Name' (Kathare), 'Aadhar Number' (412897271457), 'Mobile No.' (9511860406), 'Email' (shrinivas.22110726@viit.ac.in), and 'Password' (masked with dots). A red 'Register' button is below the password field. Below the button, a link says 'Already have an account? Click here to login'. At the bottom, a red pill-shaped box displays 'Your Account: 0xad1a20f668a98cccefff10aabf3aec07c808a226'.

Fig Sign-in

The screenshot shows the 'Login to an existing account' form on the BLOCKCHAIN-VOTE website. The layout is similar to the registration page. The main heading is 'Login to an existing account'. The form fields are 'Aadhar Number' (shrinivas.22110726@viit.ac.in) and 'Password' (masked with dots). A red 'Login' button is positioned below the password field. At the bottom, a red pill-shaped box shows 'Your Account: 0xad1a20f668a98cccefff10aabf3aec07c808a226'.

Fig Login



Fig About Page

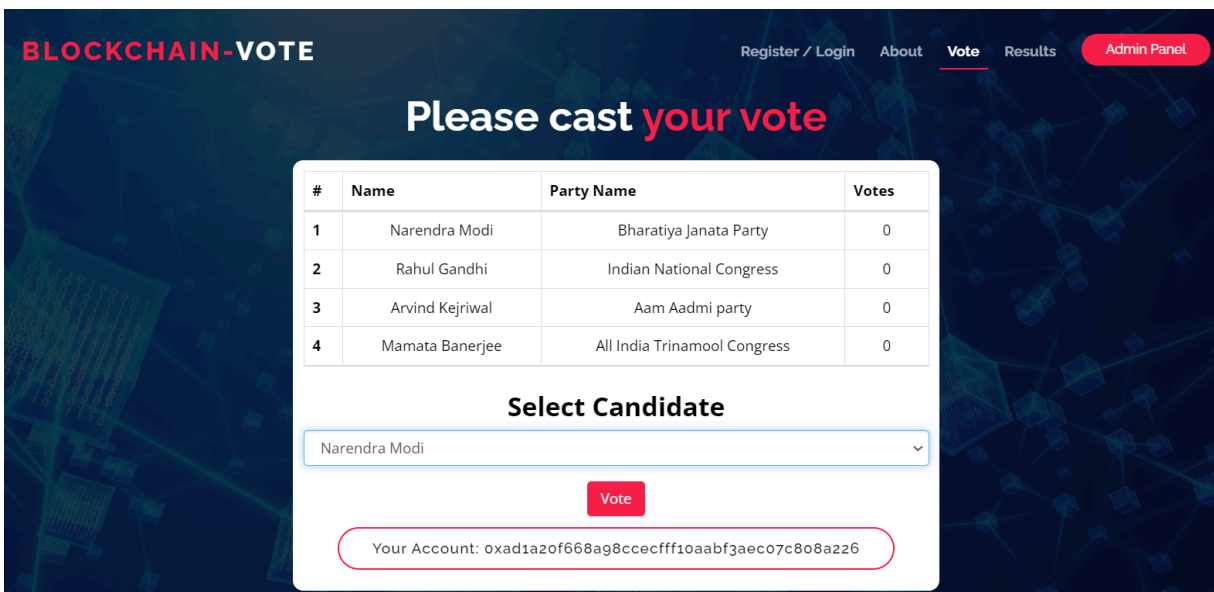


Fig Vote Page

BLOCKCHAIN-VOTE

Start Election

End Election

Candidates

Register / Login

About

Vote

Results

Admin Panel

#	Name	Party Name	Votes
1	Narendra Modi	Bharatiya Janata Party	1
2	Rahul Gandhi	Indian National Congress	1
3	Arvind Kejriwal	Aam Aadmi party	1
4	Mamata Banerjee	All India Trinamool Congress	0

Add new Candidate

Voters

Name	Aadhar Number	Email	Address
Shrinivas Kathare	412897271457	shrinivas.22110726@viit.ac.in	0xad1a20f668a98ccecfff10aabf3aec07c808a226
Shreyas Raijade	417897271123	shreyas.22110031@viit.ac.in	0x66e8490696f99494ef61de76b52b64cc292ffa6e
Tanish Bagal	432897271789	Tanish.22110655@viit.ac.in	0x151234281aa84a537bd49a4b1daeb179a5ae4158
Harshal Dahiwal	499897271321	harshal.22110863@viit.ac.in	0x501a95eb6e3175b14c7efd6511d311dc3c0e1330

Your Account: 0xad1a20f668a98ccecfff10aabf3aec07c808a226

Fig.Admin Panel

BLOCKCHAIN-VOTE

Register / Login

About

Vote

Results

Admin Panel

Election Results

#	Name	Party Name	Votes
1	Narendra Modi	Bharatiya Janata Party	1
2	Rahul Gandhi	Indian National Congress	1
3	Arvind Kejriwal	Aam Aadmi party	1
4	Mamata Banerjee	All India Trinamool Congress	0

Your Account: 0xad1a20f668a98ccecfff10aabf3aec07c808a226

Fig Result Page

10. Discussion and Conclusion :

With the blockchain-based electronic voting system, voting has significantly advanced, resolving many of the issues with conventional voting procedures. This solution makes sure that the election process is trustworthy and dependable by utilizing the transparent, secure, and immutable properties of blockchain technology. The thorough process described in the project has shown that putting such a system in place may significantly increase voter turnout, reduce the possibility of fraud, and speed up the results while keeping everyone safe. The several test cases that are run verify that the system meets both functional and non-functional criteria and operates as intended. All things considered, the implementation of a blockchain-based electronic voting system has the potential to completely transform the electoral process, increasing public trust in the democratic process and opening the door for more creative approaches to future voting systems. As technology develops further, incorporating blockchain technology into voting procedures is essential to creating an election environment that is more trustworthy, efficient, and inclusive.